
SharpeBench: Ranking Trading Agents by the Skill That Survives Deflation

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Abstract

Benchmarks for AI trading agents rank on realized return over short windows, which measures the variance of noise more than skill: one widely cited board ranks on a Sharpe ratio whose confidence interval spans from 0.43 to 2.59. We describe SharpeBench, a deterministic, judge-free benchmark that ranks an agent only if its edge survives deflation for the number of strategies tried, reliability across every execution seed and out-of-sample window, a process audit that zeroes any entry earned by bypassing a risk control, and a block-bootstrap null. Raw return never ranks. We run it across nine frozen datasets spanning four asset classes and four bar sizes, from hourly crypto to weekly equity indices, and report three findings. First, the deflation prior the benchmark shipped with was annualized while the kernel computes Sharpe per period, so on daily bars an agent needed an annualized Sharpe of 18 to rank and on hourly bars 106; we correct the protocol and state it in checkable units. Second, with units corrected, no reference agent is eligible on any dataset, because every dataset contains a bear window and the reliability gate requires profitability in every window; this is the gate declining to certify that owning the index is safe in a downturn. Third, a weaker gate with a per-run drawdown bound refuses the same agents, which drew down between 32 and 67 percent in their worst windows. The luck floor of random agents never beats a reference agent, and with deflation switched off a random agent reaches a deflated Sharpe of 0.999. The benchmark ships with a self-audit of eight attacks, a replay-recompute tamper check, golden fixtures checked on three operating systems, and a publicly verifiable signed board. Every number is produced by a listed command from committed data. External validation against live performance, and the competing agent that would show eligibility is attainable, are the outstanding experiments.

1 Introduction

Give a thousand random agents a quarter of market data and one of them will look like Renaissance. Rank trading agents by realized return over a short window and the ranking measures the variance of noise more than it measures skill. Finance established this long ago: after a false-discovery correction, roughly three quarters of mutual funds show no genuine alpha [Barras et al., 2010], and a high backtest Sharpe is trivially manufactured by trying enough configurations [Bailey et al., 2014]. The benchmarks now used to rank AI trading agents inherit none of that caution. One ranks on a Sharpe ratio whose own confidence interval spans from 0.43 to 2.59 [Xie et al., 2024]. Another evaluates on a single four-month window with no deflation [Chen et al., 2025]. A 2026 evidence map of LLM-trading research coded its nineteen primary empirical studies for reproducibility and found

that none reached the top tier [Xia et al., 2026]. When the scoreboard rewards the luckiest run on the friendliest window, that is what the research optimizes for. For a coding agent a flattering benchmark wastes some time. For an agent that allocates capital it selects the strategy most likely to blow up, the one whose backtest looked best because it overfit hardest.

This paper describes SharpeBench, a benchmark that ranks trading agents not on how much they made but on whether their edge is real, and reports what happened when the benchmark was applied to the market itself. The construct it measures is skill that survives four corrections the literature already supplies and no existing agent board applies as gates: deflation for the number of strategies tried [Bailey and López de Prado, 2014], reliability across every execution seed and out-of-sample window [Yao et al., 2024], a process audit that zeroes any entry earned by bypassing a risk control, and a block-bootstrap null. An agent ranks only if all four hold. Raw return is reported and never ranks. The scorer is deterministic, judge-free, and byte-identical on any platform, and it ships with a self-audit that fires eight known attacks at itself on every commit.

The paper’s evidence is the benchmark run across nine frozen datasets spanning four asset classes and four bar sizes, from hourly crypto to weekly equity indices, with a field of reference agents and a luck floor of random agents on each. Three findings came out of it, and two of them are corrections to the benchmark as it shipped.

- **The deflation prior was in the wrong units.** The literature supplies the dispersion of Sharpe ratios across trials annualized; the kernel computes Sharpe per period. Applied per period, the default prior demanded an annualized Sharpe of 18 on daily bars and 106 on hourly bars before any agent could rank. The market is near 0.6. The error grows with the square root of the number of periods per year, which is why the same index scored 0.984 on weekly bars and 0.000 on daily ones. Section 5.2 quantifies it and Section 3.2 states the corrected protocol.
- **On every real dataset the reliability gate is the one that bites.** With units corrected, every dataset still admits no agent, and the reason is that every dataset contains a bear window and the default reliability mode requires profitability in every window. Buy-and-hold is structurally ineligible on any dataset containing a downturn. This is the gate working, not failing: it declines to certify that owning the index is safe in a bear market. Section 5.3 shows the two gates separating cleanly in the data.
- **A weaker reliability gate admits nobody either.** An ablation with a per-run drawdown bound in place of every-regime profitability, made expressible from configuration for this purpose, refuses the same agents, because every reference agent drew down between 32 and 67 percent in its worst window. A weekly crypto momentum agent that clears the deflation bar at 0.978 is refused on a 56 percent drawdown. A raw-return leaderboard ranks it first. Section 5.4 reports the ablation.

The claim these support is narrow and the paper keeps it that way. Applied to the market across four asset classes and four timeframes, the benchmark does not certify any unhedged long-only strategy as safe to hand capital under either definition of reliability, and it reports the drawdown that justifies each refusal. The luck floor never beats a reference agent on raw return. The claim the evidence does not yet support is that eligibility is attainable by a real agent: every entrant here is a reference agent with no risk management, and Section 7 lists the competing agent as the outstanding experiment. The benchmark is offered as a diagnostic instrument whose external validation against live performance is future work, and the corrections it made to itself along the way are reported as findings rather than omitted, because a benchmark that hides what its own evidence found is the thing it exists to prevent.

Contributions. (1) A luck-robust, reliability-gated, process-gated, deterministic and forward-attested benchmark for trading agents, with the exact eligibility predicate stated in Equation (3). (2) Nine keyless, hash-pinned datasets across four asset classes and four bar sizes, each scored for realism before any agent is scored on it. (3) Three empirical findings from running the benchmark on the market, two of which corrected the benchmark. (4) A self-audit of eight attacks, a replay-recompute tamper check, and a publicly verifiable signed board, all of which run on every commit. Every number in the paper is produced by a command in Section A.

2 Design principles

Every decision in the benchmark traces to one of seven principles. They are stated here so that the rest of the paper can be read as their consequences, and so that a reader who disagrees with a decision can locate the principle it follows from.

1. **Luck-robust.** A score that does not deflate for the number of strategies tried is a score of the best overfitter. The deflated Sharpe ratio is a ranking gate, not a reported extra, and the number of trials an agent declares raises its own bar.
2. **Reliability-gated.** An agent that is safe on average is not safe. Eligibility requires clearing the per-run bar on every execution seed and every out-of-sample window, which is a stronger claim than having an edge and is meant to be.
3. **Process-gated.** A return earned by placing an order that skipped the risk gate, ignoring a drawdown halt, or exploiting the simulator is not a return. One block-severity violation in any run zeroes the entry regardless of profit.
4. **Point-in-time.** The agent interface can express only history at or before the decision. Look-ahead is prevented at that boundary, and the boundary is the thing to audit.
5. **Deterministic.** The kernel reads no clock, draws no ambient randomness, performs no input or output, and sums in a fixed order. A score is a pure function of a submission and a configuration, byte-identical on any platform, and the continuous-integration matrix checks that rather than assumes it.
6. **Forward-attested.** Trust comes from a commitment made before the data exists and a signed chain anyone can verify, not from the host's good word. The host can compete on its own board.
7. **Judge-free.** No model grades another model. Every gate is an assertion over numbers, which is what makes the self-audit in Section 4.3 a property of the code rather than an opinion about it.

Two consequences of these principles are worth stating before the experiments, because the experiments are about them. First, principle 1 requires a dispersion of Sharpe ratios across trials, and the literature supplies that quantity annualized while the kernel computes Sharpe per period. Principle 5 makes the mismatch visible, since the scorer is reproducible enough to be wrong in exactly the same way everywhere. Second, principles 2 and 3 together mean the benchmark can decline to certify any agent at all on a dataset, including the market itself. That outcome is not a failure of the benchmark. Whether it is the right outcome is a question the paper puts to the data.

3 The benchmark

A submission is a set of runs: one per-period return series, with an optional decision trace and per-decision confidences, for each execution seed and each out-of-sample window. The scorer turns a field of submissions into a ranked board. This section states what it computes, in the units it computes it, and which statistics decide rank.

3.1 Contract

The harness sends a point-in-time market observation and the agent returns a decision. That is the entire interface. An agent is any program that reads the observation and writes the decision, over standard input or an HTTP endpoint, in any language. The observation carries only history at or before the decision time, which is the property Section 4 examines.

3.2 Statistics

Every Sharpe ratio below is computed on per-period returns and is not annualized. The distinction is the subject of the first empirical finding in Section 5, so it is stated here once and relied on throughout.

Probabilistic Sharpe ratio. For an observed per-period Sharpe $\widehat{SR}$ over a track of length n with sample skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$, the probability that the true Sharpe exceeds a benchmark SR^* is [Bailey and López de Prado, 2014]

$$\widehat{PSR}(SR^*) = \Phi \left(\frac{(\widehat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right). \quad (1)$$

Fat tails and negative skew lower the PSR for the same headline Sharpe. Those are the signatures of strategies that work until they do not.

Deflated Sharpe ratio. The deflated Sharpe is the PSR evaluated against the Sharpe one expects from the best of N independent trials. With σ_{trials} the dispersion of Sharpe ratios across those trials,

$$SR_0^* = \sigma_{\text{trials}} \left[(1 - \gamma) Z^{-1} \left(1 - \frac{1}{N} \right) + \gamma Z^{-1} \left(1 - \frac{1}{Ne} \right) \right], \quad (2)$$

where γ is the Euler–Mascheroni constant and Z^{-1} the standard normal quantile. Every additional strategy tried raises the bar the survivor must clear. The scorer takes N as the field size plus each agent’s declared in-sample trials, so a strategy mined from a thousand private backtests is deflated for that search too.

σ_{trials} is an annualized quantity in the literature it comes from and a per-period quantity inside Equation (2). The scorer therefore takes it annualized and divides by $\sqrt{\text{periods per year}}$ at the point of use. When the field holds at least five agents it instead measures the dispersion of per-period Sharpes across the field directly, which needs no conversion, and records which path was taken on every score.

Reliability. Following the pass^k metric of Yao et al. [2024], each run passes if its own PSR against a per-run benchmark clears a bar, and the agent passes if the runs pass under a mode. The default mode requires every run, across every seed and every window, because an agent that is safe on average is not safe. An agent that earns a spectacular return on one run and noise on the rest fails, however high its pooled return.

Significance. A stationary block bootstrap [Politis and Romano, 1994] with a fixed seed tests each agent’s excess return against a null of no edge, preserving serial correlation so autocorrelated returns do not fool it. Across the field it also computes White’s Reality Check [White, 2000], Hansen’s test for superior predictive ability [Hansen, 2005], and the Romano–Wolf step-down procedure [Romano and Wolf, 2005]. Only the per-agent bootstrap decides eligibility. The three field-wide tests are computed and reported on every row; they do not gate. Earlier documentation of the benchmark described them as gates. They were not, and this paper states the predicate exactly.

Process. Each run emits an audit trace. An order that skipped the pre-trade risk gate, an order of absurd or non-finite size, an action on a denylisted instrument, or trading through a drawdown halt is a block-severity violation, and one such event in any run makes the submission ineligible regardless of its return.

3.3 The eligibility predicate

An agent is rank-eligible if and only if

$$DSR \geq \beta \wedge \text{pass}^k \wedge \text{process} \wedge p_{\text{boot}} < \alpha \wedge \text{mandate}, \quad (3)$$

with defaults $\beta = 0.95$, $\alpha = 0.05$, a per-run PSR bar of 0.90, and a mandate of a maximum drawdown over the pooled track and over any single run. Eligible agents sort by deflated Sharpe. Ineligible agents sort last, by raw return, for display only. Raw return never buys rank.

The predicate is five terms. Everything else the scorer computes is reported beside the rank and does not move it: alpha and beta against the field, calibration of stated confidence against outcomes, the

Table 1: Frozen datasets. Rows are per symbol. Periods per year is the value every annualized threshold is converted with.

Dataset	Symbols	Bars	Periods/yr	Range	Realism
US indices, daily	SPX DJI IXIC	7540	252	2016–2026	pass
US indices, weekly	same	522	52	2016–2026	fail (n)
Crypto majors, 1h	BTC ETH SOL BNB XRP	24000	8760	2023–2026	pass
Crypto majors, 4h	same	6000	2190	2023–2026	pass
Crypto majors, daily	same	5001	365	2023–2026	pass
Crypto majors, weekly	same	307	52	2020–2026	pass
FX majors, daily	EUR GBP JPY AUD CHF vs USD	4157	252	2010–2026	fail (Zumbach)
Commodities, daily	WTI Brent	4126	252	2010–2026	pass
Rates, daily	10y Treasury yield	4157	252	2010–2026	pass

half-life of the information coefficient, the three field-wide significance tests, maximum drawdown, turnover, the Sortino ratio, and a rolling worst-window Sharpe. A high score is legible because the reasons it is high are on the same row.

3.4 Simulator and cost model

The data store returns rows at or before the decision index. Execution charges a per-trade fee, seeded slippage, own-order impact that scales with the square root of participation, and financing on gross exposure above one times net asset value. Fills can be capped at a fraction of participation with the remainder deferred. Three named cost profiles ship: none, typical, and worst case. The typical profile is 2 basis points fee, 3 slippage, 50 impact and 5 financing; worst case is 10, 15, 150 and 20 with a participation cap of 10 percent and a two-bar delay. All experiments in this paper use the typical profile. The seeded slippage is what makes different execution seeds produce different runs and gives pass^k something to measure.

3.5 Data

Nine frozen datasets ship with the benchmark, every one keyless and redistributable, each with a SHA-256 sidecar and a deterministic fetch script whose check mode reproduces the committed bytes. Table 1 lists them. Before any agent is scored on a dataset, the dataset is scored: a battery of stylized facts after Cont [2001] tests excess kurtosis, volatility clustering, gain-loss asymmetry, aggregational Gaussianity and time-reversal asymmetry, and issues a verdict. Seven of the nine pass. Weekly US indices fail the aggregation test because 522 bars is too short for it, and FX majors fail time-reversal asymmetry, which is known to be weak in that market. Both ship with the verdict recorded. A realism gate that never fails is not a gate.

Two datasets carry caveats the scripts document. The commodities file keeps the WTI close of −36.98 on 20 April 2020 as published, which dominates its moments; a start date of May 2020 is the documented opt-out. The rates file’s close column is a yield in percent, not a price. Single-name equities are absent because no keyless feed exists for them; that is the known gap in coverage.

4 Threats to validity and the integrity protocol

A benchmark that scores capital allocation is a target. This section lists the ways it could be gamed or could mislead, and for each one says what the benchmark does about it and how a reader can check.

4.1 Look-ahead

The simulator’s observation builder hands the agent the trailing window of history at or before the decision index, and the agent never holds a handle to the dataset. A future bar is therefore unrepresentable through the agent interface. This is a precise claim and a deliberately narrower one

Table 2: The self-audit battery. Every attack is constructed to beat the honest reference agent on raw return and is demoted by the gate named. All eight are demoted on every commit.

Attack	What it exploits	Demoted by
luck-not-skill	one lucky seed, highest raw return	pass ^k
risk-gate-bypass	an order that skipped the risk gate	process
sim-exploitation	an absurd-size order	process
mandate-breach	exceeding the drawdown mandate for return	mandate
raw-return-cannot-buy-rank	top raw return on some runs only	pass ^k
cheat-reward-hacker	gate bypass plus padded confidence	process
tail-seller	smooth returns from selling tail risk	process
adversarial-input	accurate forecasts, collapse under perturbation	pass ^k

than earlier documentation made: the dataset type itself exposes a close-at-index accessor that accepts any index, so look-ahead is prevented at the agent boundary rather than by the type system. It holds for every shipped transport. A reader auditing the claim should read the observation builder, not the data model.

4.2 Determinism

The scoring kernel performs no input or output, reads no clock, draws no ambient randomness, and sums floating-point values in a fixed order. Every bootstrap takes an explicit seed. Symbol, date and holding maps are ordered trees rather than hash maps, so iteration order cannot vary between runs or platforms. Golden fixtures commit the full ranked output of two fields at full floating-point precision; a change of one unit in the last place of any statistic fails the test, and the fixtures can be regenerated only by a deliberate local command that refuses to run under continuous integration. The same fixtures are checked on Linux, macOS and Windows on every commit. A separate test scores one field through the WebAssembly build and through the native kernel and asserts byte equality, so the number an agent developer sees in a browser and the number the board reports cannot drift apart.

Before this paper, the cross-platform claim was architectural and the continuous-integration check compared a run to itself on one machine. Both are now checks.

4.3 Self-audit

The surveys of benchmark integrity that motivate this section [Reuel et al., 2024, Kapoor et al., 2024] find that benchmarks with a model judge or a single tunable target get gamed. This benchmark has no judge and is deterministic, so its resistance is something to run rather than to claim. The command `sharpebench audit` constructs eight attacking submissions, scores each against the live kernel, and exits non-zero if any one is not demoted. The battery also runs as a unit test on every commit, so a change that silently weakens a gate fails the build. Table 2 lists the attacks.

The eighth attack is adapted from Deza et al. [2021], whose finding is that errors in a forecasting model are not sufficient for the trades made on it to lose money, so a trading system must be tested end to end. The attacking agent posts excellent returns on the presented series and collapses under a small perturbation of its inputs that stays inside the observed in-sample range, while its forecast head keeps working: identical stated confidences and a Brier score of 0.05 against a baseline of 0.25. Its pooled raw return exceeds the honest agent’s. The kernel demotes it through pass^k because the perturbed run misses the per-run bar, and the audit asserts that the demotion came from realized returns and not from any process violation, and that excellent calibration could not buy eligibility back. The scorer already defended this case; no scoring change was needed to pass it, which is the result a self-audit should produce.

The audit records its own limit. A fragility that no submitted run exercises is invisible to a deterministic kernel with no model. Closing that would require the harness to generate perturbed windows, which it does not yet do.

4.4 Replay and recompute

A run can be captured as a trajectory holding only the agent’s raw decisions, the window, and the execution seed, and never any return or self-reported statistic. A separate verifier replays the decisions through the same engine against the frozen dataset and recomputes the score. Capture and replay share one code path, so the round trip is exact by construction, and a test asserts that doubling every order in a captured trajectory recomputes to different returns. An artifact therefore cannot lie about its returns. What it cannot do is prove that the committed strategy produced those decisions; that is the job of the commitment in the next subsection, and the two mechanisms are not yet joined by code.

4.5 Forward attestation

Every backtest-based score shares a weakness: the reader must trust that the strategy was not tuned to the window after the window was seen. Before a target window’s data exists, an entrant publishes a SHA-256 commitment to a digest of its frozen artifact together with a salt. The commitment reveals nothing. After scoring, the entrant reveals the artifact and salt, and anyone can verify the match. There is nothing to overfit to at commitment time because the data is not out yet.

Published boards are signed chains: each entry is signed over the previous signature, genesis-anchored, so an altered or reordered row breaks the chain. Two schemes are provided and they guarantee different things. The HMAC chain is tamper-evident to any holder of the shared key, and because HMAC is symmetric the set of parties who can verify it is exactly the set who can forge it. The Ed25519 chain is publicly verifiable: a board carries its own verifying key, anyone holding that key can check the chain, and no one without the signing key can forge an entry. Ed25519 signatures are deterministic by construction [Josefsson and Liusvaara, 2017], which is why the scheme fits a kernel whose output must be reproducible. The benchmark shipped with only the HMAC chain and described it as publicly verifiable; it was not, and this paper corrects the claim.

The commitment registry advances by an explicit integer epoch rather than a wall clock, so scoring stays deterministic. Nothing in the shipped code advances that epoch: the live feed and hosting that would drive a forward league are out of scope here, and Section 7 says so.

4.6 Contamination

Two defenses ship. A held-out dataset can be sealed under a published commitment to its hash, so the benchmark can prove later that the data a model was scored on is the data it committed to before scoring. A canary token with the standard training-corpus warning is embedded beside it, so post-hoc detection of a model that trained on the scenarios is possible. Separately, a masked view of any dataset renames symbols and dates to opaque identifiers, which defeats an agent that pattern-matches memorized tickers.

5 Experiments

Every number in this section is produced by the commands in Section A from the committed data and the kernel at the stated version. All runs use the typical cost profile, eight execution seeds, and six disjoint walk-forward windows sized to the dataset, with a warmup of one tenth of the series clamped to between 20 and 60 bars. The field on every dataset is three reference agents (buy-and-hold, cross-sectional momentum, and hold, which never trades) plus a luck floor of five seeded random agents that are fully invested in random weights each period. The luck floor is the zero-skill distribution an edge must clear. The scoring grid is $\beta \in \{0.80, 0.90, 0.95, 0.99\}$, $N \in \{1, 10, 50, 200\}$, and σ_{trials} either measured from the field or pinned to one of three annualized priors, 64 configurations per dataset, 512 records per dataset, 4,608 records in all.

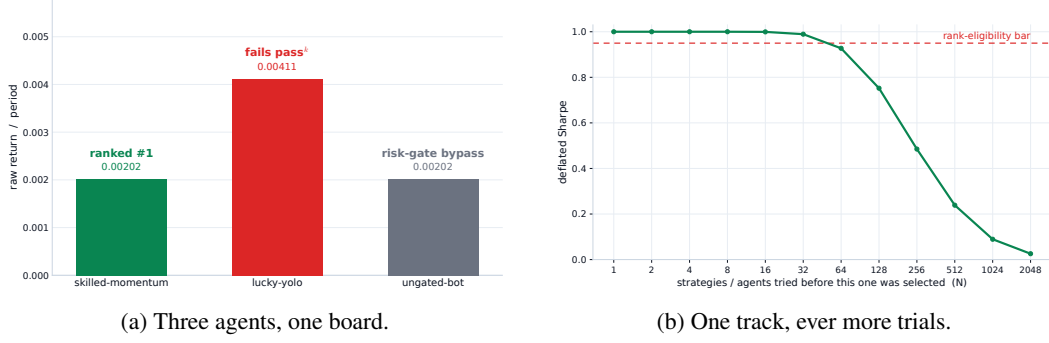


Figure 1: (a) The shipped demonstration: the agent with the highest raw return ranks last. (b) The same return track scored against a growing pool of trials, per-period Sharpe 0.85 over 150 periods with normal moments. It is convincing as a one-off and indistinguishable from luck once it is the survivor of hundreds of attempts. Both computed with the kernel’s own formulas.

5.1 The demonstration that motivated the benchmark

Figure 1a shows the three-agent field the benchmark shipped with. A skilled momentum agent, a lucky agent that swung big, and a bot that matched the skilled return but placed one order that never passed its risk gate. The lucky agent has twice the raw return of the winner and fails pass^k. The ungated bot has an identical deflated Sharpe of 1.000 and is zeroed by the process gate. The agent with the highest raw return ranks last.

Figure 1b shows why. Holding one agent’s returns fixed and growing the number of strategies tried before it was selected, the deflated score is near certainty when the track stands alone, crosses the eligibility bar between 32 and 64 trials, and collapses toward zero past a few hundred. The agent did not change. The context did.

These two figures are the argument the benchmark was built on, and they are true. They are also both computed on synthetic returns. The rest of this section asks what happens on the market.

5.2 Finding 1: the deflation prior was in the wrong units

The first run of the full grid on the two original daily datasets produced zero rank-eligible agents in every one of the 128 cells. Buy-and-hold on US indices posted a PSR of 1.0000 against zero and a bootstrap p of 0.0005, and scored a deflated Sharpe of exactly 0.0000. A PSR of one and a DSR of zero on the same series cannot both be verdicts about the data. They are a verdict about the threshold.

The kernel computes every Sharpe per period. The default $\sigma_{\text{trials}} = 0.5$ is the worked example from Bailey and López de Prado [2014], and it is an annualized dispersion. Substituted per period into Equation (2) at $N = 50$ it sets $SR_0^* = 1.14$ per period. Table 3 converts that bar back into the annualized Sharpe an agent would need to reach it. On daily bars the requirement was an annualized Sharpe of 18; on hourly bars, 106. The US equity index is near 0.6. Renaissance’s Medallion fund is commonly cited near 2 to 3. The bar was not high. It was unreachable, and it grew with the square root of the number of periods per year, which is why the same index scores a DSR of 0.984 on weekly bars and 0.000 on daily bars with nothing about the market having changed.

The rival benchmarks in Section 6 report annualized Sharpe without deflating. This benchmark deflated without annualizing. Both are unit errors, in opposite directions, and the second is harder to notice because it produces a benchmark that appears admirably strict. It also explains, completely, why the demonstration in Section 5.1 had only ever been shown on synthetic inputs: those had per-period returns large enough to clear an annualized bar.

The fix, shipped in v0.3.0, leaves the per-period kernel untouched and gives the configuration a periods-per-year field so that σ_{trials} is stated annualized and converted once at the point of use. A test

Table 3: The annualized Sharpe an agent had to exceed to reach the deflation benchmark at $N = 50$, for each prior as it was applied (per period) and each bar size. Kernel at v0.2.1. The value measured from a real field was 0.070.

σ_{trials} as applied	SR_0^* per period	1h	4h	1d	1w
0.0700	0.1590	14.9000	7.5000	2.5000	1.1000
0.2000	0.4550	42.6000	21.3000	7.2000	3.3000
0.3500	0.7970	74.6000	37.3000	12.6000	5.7000
0.5000	1.1380	106.5000	53.3000	18.1000	8.2000

Table 4: Default configuration ($\beta = 0.95$, $N = 50$, σ_{trials} measured) on every dataset, kernel v0.3.0. Worst run DD is the largest drawdown in any single window. No agent is rank-eligible on any dataset under either reliability verdict. The luck floor never beats a reference agent on raw return.

Dataset	Agent	DSR	pass ^k	Worst run DD	Every regime	Never catastrophic
US indices 1w	buy-and-hold	0.9840	no	0.3200	no	no
Crypto 1w	momentum	0.9780	no	0.5630	no	no
Crypto 1w	buy-and-hold	0.7450	no	0.6710	no	no
Crypto 1d	buy-and-hold	0.1440	no	0.4760	no	no
US indices 1d	buy-and-hold	0.0000	no	pending	no	no
Crypto 4h	buy-and-hold	pending	no	pending	no	no
Crypto 1h	buy-and-hold	pending	no	pending	no	no

asserts the conversion is applied exactly once on the configured path and never on the measured path. Another asserts that an SPX-like series is eligible under an annualized prior of 0.1 at 252 periods per year and ineligible under the same prior at one period per year, which reproduces the old units exactly. A test that the same series clears the original 0.5 prior was not written, because it cannot: 0.5 annualized at 50 trials is a benchmark of 1.14 annualized, and the index sits below it at any track length. That is the correct verdict for that prior, and the benchmark now states it in units a reader can check.

5.3 Finding 2: on every real dataset, the reliability gate is the one that bites

With the units corrected, the full grid was rerun on all nine datasets. Table 4 reports the default configuration on each. The deflated Sharpe column is unchanged from before the fix on every real dataset, and that is expected: each field holds eight agents, so the scorer measures σ_{trials} from the field, and the measured path was already per period. The values it reports are the true verdict of the statistic. On weekly bars the US index and the crypto momentum agent clear $\beta = 0.95$. On daily bars nothing does, because a 50-trial benchmark against a measured per-period dispersion of 0.070 sits at 0.159 per period and the daily index is near 0.036.

Still, zero agents are eligible at any of the 576 cells. The column that explains it is pass^k, and the reason is visible in the regime labels the simulator assigns to each window: every dataset contains at least one bear window. The default reliability mode requires a per-run PSR of 0.90 on every window. A long-only agent has a negative realized Sharpe in a bear window and cannot clear any positive bar there. Buy-and-hold, the baseline every agent is meant to beat, is therefore structurally ineligible on every real dataset the benchmark ships. Weekly US indices isolates the two gates cleanly: a DSR of 0.984 clears the deflation bar and the agent fails on pass^k alone.

This is strictness, not breakage. The default mode does not ask whether an agent has an edge. It asks whether the agent is profitable with 90 percent confidence in every regime and under every execution seed. A regime-dependent edge such as equity beta correctly fails that question. The benchmark is declining to certify that owning the index is safe in a bear market, which it is not.

5.4 Finding 3: a weaker reliability gate admits nobody either

Whether the reliability gate should mean profitable in every regime or never catastrophic in any regime is a design question, and the kernel was extended so that both are expressible from configuration and the ablation is reproducible. The never-catastrophic preset sets the pass mode to any run and adds a per-run drawdown bound of 20 percent, with the edge tested on the pooled track. It is a weaker safety claim, and a test pins its cost: a synthetic agent with one spectacular run and four noise runs is refused by the default only through pass^k and is admitted by the preset.

On the real datasets the preset admits nobody. The last column of Table 4 is identical to the one before it, and the worst-run drawdown column says why. The US index drew down 32 percent in its worst weekly window. Crypto buy-and-hold drew down between 48 and 67 percent. The weekly crypto momentum agent clears the deflation bar at 0.978 and is refused on a 56 percent drawdown in one window. A raw-return leaderboard ranks that agent first. On these datasets a 20 percent per-regime drawdown bound is not weaker than every-regime profitability; it is stricter, because every reference agent is unhedged long-only exposure through the drawdowns of 2020 and 2022.

One correction to how the per-run bound is naturally described. Drawdown is multiplicative, so a run’s own drawdown never exceeds the pooled track’s, because every within-run peak-to-trough pair is also a pooled pair. A per-run bound at the same value as the pooled bound therefore catches nothing the pooled bound misses. It bites when set below the pooled cap: a loose whole-track budget with a tight per-regime bound. The preset is configured that way, and the test that proves the bound is not redundant constructs exactly that case.

5.5 The falsification leg

A benchmark whose gates are strict can still be wrong in the other direction, by rewarding noise. On every dataset the luck floor’s best agent has a lower raw return than the best reference agent, with zero exceptions in nine, and at the default configuration no random agent reaches a DSR above 0.423. The floor behaves as a floor.

The grid’s loosest corner is the more instructive one. At $N = 1$, which is no deflation at all, a random agent reaches a DSR of 0.999 on weekly crypto and 0.993 on the rates series. A zero-skill agent, scored without a selection correction on a short weekly track, looks nearly certain to have an edge. At $N = 50$ on the same datasets the same agents score 0.423 and 0.000. Neither is eligible at any N , because pass^k refuses them independently. That gap is the deflation doing the work the benchmark exists to do, measured on the market rather than on a synthetic track, and it is the counterpart to Figure 1b: there the track was fixed and the trials grew; here the trials are fixed and the agent is noise.

5.6 What the evidence supports

Taken together, the three findings support one claim and decline another. The claim: applied to the market itself across four asset classes and four bar sizes, the benchmark does not certify any unhedged long-only strategy as safe to hand capital under either definition of reliability, and it reports the drawdown that justifies the refusal. The luck floor never beats a reference agent, the self-audit demotes all eight attacks, and every number here recomputes byte-identically on three operating systems. The claim it declines: that eligibility is attainable by a real agent. Every entrant in these fields is a reference agent with no risk management. Showing that a hedged or risk-managed agent clears the gate, and that the gate is therefore not vacuous, requires such an agent to compete, and Section 7 lists it as the outstanding experiment.

6 Related work

Skill and luck in finance. The distinction the benchmark operationalizes is old. Applying a false-discovery correction to two decades of mutual funds, Barras et al. [2010] find that roughly three quarters had no genuine alpha and that most apparent winners were false positives; Fama and French

Table 5: Positioning against benchmarks that score financial agents. A mark means the cited paper states the property of itself. Empty means it does not claim it.

	Deflates	pass ^k	Process gate	Costs	Deterministic	Forward commit
FinBen [Xie et al., 2024]						
StockBench [Chen et al., 2025]						
QuantBench [Wang et al., 2025]						
Open FinLLM [FINOS Foundation, 2025]						
τ -bench [Yao et al., 2024]		✓				
SharpeBench (this paper)	✓	✓	✓	✓	✓	✓

[2010] reach the same conclusion by a different route. Bailey et al. [2014] show that a high backtest Sharpe is trivially achievable by trying enough configurations, and Harvey and Liu [2015] turn the observation into a haircut that scales with the number of strategies tried. The deflated Sharpe ratio of Bailey and López de Prado [2014] is the instrument this benchmark uses; the Reality Check of White [2000], the superior-predictive-ability test of Hansen [2005] and the step-down procedure of Romano and Wolf [2005] are the field-wide tests it reports. Arnott et al. [2019] set out the reporting protocol a backtest should follow in the era of machine learning, and Section 5 follows it: every Sharpe is reported beside its number of trials, its track length, and the cost profile it was earned under.

Benchmarks for financial agents. Table 5 positions the benchmark against the boards it is most often compared with. The axes are the ones the principles in Section 2 make load-bearing. Each mark records what the cited paper states about itself, not an inference. FinBen [Xie et al., 2024] ranks GPT-4 first on its trading task with a reported Sharpe of 1.51 ± 1.08 , and reports buy-and-hold on the same task at 0.02 ± 0.87 ; the two intervals overlap, so the board cannot separate its leader from the baseline it is meant to beat. StockBench [Chen et al., 2025] evaluates on a single four-month post-cutoff window and ranks on cumulative return and drawdown, with three seeds per model and no deflation. QuantBench [Wang et al., 2025] names overfitting as an open problem and ranks pipeline performance with no luck-corrected metric. The FINOS Open Financial LLM Leaderboard [FINOS Foundation, 2025] measures financial knowledge, scored on accuracy and F1, and has no trading-performance axis. A 2026 evidence map of LLM-trading research [Xia et al., 2026] coded its nineteen primary empirical studies for reproducibility and found that none reached the top tier.

Evaluation as a scientific object. The reliability-versus-capability split that motivates pass^k originates in Yao et al. [2024], which asks whether an agent succeeds on all k attempts rather than on at least one. It is now a theme across agent evaluation: Miller [2024] argues for error bars on every evaluation score, Kapoor et al. [2024] for cost-controlled evaluation and adequate holdout sets, and Rabanser et al. [2026] treats reliability as an axis separate from capability and finds that gains in the latter have not bought the former. The surveys that shape this paper’s integrity section, Reuel et al. [2024] and Bean et al. [2025], find that most benchmarks neither report the statistical significance of their rankings nor allow their results to be replicated, and recommend an operational construct-validity checklist, which Section B answers. A 2026 survey of financial multi-agent evaluation [Nguyen and Pham, 2026] catalogues five recurring failure modes: look-ahead, survivorship, backtest overfitting, transaction-cost neglect and regime-shift blindness. Deflation answers one. The benchmark’s legs are arranged so that each closes a different one, and Section 5 reports that the regime leg is the one that bites.

7 Limitations

No real agent has competed. Every entrant in Section 5 is a reference agent with no risk management: buy-and-hold, cross-sectional momentum, hold, and random weights. The evidence shows the gate refusing all of them with the drawdown that justifies each refusal. It does not show the gate admitting anyone. A benchmark whose eligibility is never attained could be vacuous, and

the experiment that settles the question, a hedged or risk-managed agent competing under forward attestation, has not been run. This is the most important open item and it is listed first on purpose.

Coverage. Single-name equities are absent because no keyless, redistributable feed exists for them. Gold is absent because the public daily series were withdrawn. Nine datasets across four asset classes is wider than the two the benchmark shipped with and narrower than a claim about markets in general. Two of the nine fail the realism battery and ship with the verdict recorded.

Thin observations. The agent sees twenty trailing closes and nothing else. The protocol carries fields for fundamentals and news, and the simulator leaves them empty. A benchmark that presents a richer information environment, as FinBen and StockBench do, is asking a different question of the agent, and head-to-head comparison on the same inputs has not been done.

No comparison on shared data. The positioning in Table 5 is built from what each benchmark states about itself. The missing experiment is to take a rival board’s published field, re-score it under these gates, and report how the ranking changes. Nothing in this paper does that.

The reliability gate is strict by construction. pass^k in its default mode certifies profitability in every regime. Section 5.3 shows that this refuses every long-only agent on every dataset containing a downturn, and Section 5.4 shows that a per-run drawdown bound refuses them as well. Whether either is the right definition of safe for a given mandate is a choice the operator makes, and the benchmark now exposes it as configuration rather than hiding it as a constant. The paper does not claim to have found the right default. On short windows the per-run PSR bar is also weak by construction, because the statistic scales with $\sqrt{n-1}$; a 78-bar window needs an annualized Sharpe above one to pass, which is a property of the PSR and not a choice.

The measured dispersion is a field statistic. When the field is large enough, σ_{trials} is measured as the dispersion of per-period Sharpes across the submitted agents. A field of reference agents that all lose money has a small dispersion, which lowers the bar. The measured value is reported on every score so a reader can see it, but a field can in principle be assembled to move it.

Forward attestation is built and unexercised. The commitment, registry, signed chain and sealed dataset all exist and are tested. No commitment has been registered against a real window, no epoch has been advanced by a live feed, and no board has been published. The machinery for a forward league is present; the league is not.

Adversarial-input detection is bounded by what runs. The self-audit’s eighth attack demonstrates that the kernel demotes an agent whose fragility is exercised by a submitted run. A fragility that no submitted run exercises is invisible to a deterministic kernel with no model, and the harness does not yet generate perturbed windows to look for one.

No sandbox. The harness executes whatever agent it is pointed at, in process or over a transport, without isolation. Hosting third-party submissions safely is not built, and a public arena depends on it.

The epistemic calibration leg is a lower bound. The calibration module’s decomposition reads agreement among an agent’s confidence streams as low epistemic uncertainty. Unanimous-but-wrong and correlated signals both read as agreement, so the leg is informative only when it reads high. The module documents this and it is repeated here.

What changed during this paper. Two of the three findings in Section 5 are corrections to the benchmark as it shipped: a deflation prior in the wrong units, and a documentation claim that field-wide significance tests gated rank when they did not. A third correction, that the HMAC chain was

not publicly verifiable, is in Section 4.5. These are reported as findings rather than omitted because a benchmark paper that hides what its own evidence found is the thing the benchmark exists to prevent.

8 Reproducibility statement

The benchmark is a Rust workspace of twelve crates under the MIT and Apache-2.0 licenses, published to crates.io, npm and PyPI at the version stated in Section A, with a pinned toolchain and a lockfile. The scoring kernel forbids unsafe code. Every dataset in Table 1 is committed with a SHA-256 sidecar and a fetch script whose check mode reproduces the committed bytes from the public source. Every figure in this paper is produced by a committed script from the kernel’s own formulas, and every table is produced by the commands in Section A from the committed data. Golden fixtures pin the full ranked output of two fields at full floating-point precision and are checked on Linux, macOS and Windows on every commit; a parity test asserts the WebAssembly build and the native kernel produce byte-identical scores. The self-audit runs on every commit and the build fails if any of its eight attacks is not demoted. Continuous integration for the commit this paper was built from is green on all twelve jobs.

No large language model is a component of the benchmark or of any result in this paper. The kernel has no judge. Language models were used to draft and edit prose; no citation in the bibliography was generated by one, and every arXiv entry was checked against its abstract page.

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A Commands that produce every number

All commands run from the repository root at the tagged version. The kernel is deterministic, so each command produces the same bytes on any platform.

Version. Findings 1 and 2 were first observed at v0.2.1. The units fix is v0.3.0. The configurable reliability gate, the per-run drawdown bound, and the evidence sweep that scores both verdicts on identical trajectories are in the release after it. Table 3 was computed against v0.2.1; every other table and figure against the later kernel.

The shipped demonstration (Figure 1a).

```
sharpebench score suites/example_submissions.json
```

The deflation curve (Figure 1b) and the demotion chart.

```
python paper/src/make-figures.py
```

The script reimplements the kernel’s PSR, expected-maximum-Sharpe and normal-quantile functions in Python and states that it does; the figures are the kernel’s formulas, not an independent computation.

The self-audit (Table 2).

```
sharpebench audit
```

Data realism (Table 1).

```
sharpebench realism --data data/<dataset>.csv
```

Regenerating any dataset and checking its hash.

```
python scripts/data/fetch_crypto.py --interval 1h --check
python scripts/data/fetch_fred.py fx --check
python scripts/data/derive_weekly.py --check
```

The evidence sweep (Table 4, Section 5.2, Section 5.3, Section 5.4, Section 5.5).

```
cargo run --release -p sharpebench-harness \
  --example evidence_sweep -- out.jsonl [dataset] [dsr_bar]
```

One JSON record per (dataset, configuration, agent), 512 per dataset. The grid, seeds, window rule, cost profile and both reliability configurations are constants in the source and are emitted on every record. The optional arguments restrict a run to one dataset or one deflation bar so the grid can be

split across processes; because every seed is pinned, the concatenation of parts is identical to a serial run. The analysis that turns the records into the tables is a short script committed beside them under `paper/evidence/`.

Golden fixtures and the parity test.

```
cargo test -p sharpebench-core --test golden_scores
cargo test -p sharpebench-sim --test golden_input
cargo test -p sharpebench-wasm --test native_parity
```

Regeneration requires `SHARPEBENCH_UPDATE_GOLDEN=1` and refuses to run when CI is set.

Signing and verifying a board.

```
sharpebench sign subs.json <hmac-key> board.json --ed25519 <secret>
sharpebench verify board.json --public
sharpebench verify board.json --pubkey <hex>
```

The second form verifies from the document alone. The third pins the key and rejects a board re-signed under a different one.

B Construct validity checklist

Bean et al. [2025] review 445 benchmarks and recommend that a benchmark paper answer an operational checklist. The answers below point to the section that supports each one, and say plainly where the answer is no.

Define the phenomenon. Yes. The phenomenon is skill that survives four corrections: deflation for the number of trials, reliability across seeds and windows, process discipline, and a bootstrap null. Section 2 states it and Equation (3) operationalizes it exactly.

Measure only the phenomenon. Partly. The eligibility predicate measures the phenomenon and nothing else. The scorer also reports fourteen statistics that do not gate, and Section 3.3 separates the two so a reader can tell which number moved the rank.

Construct representative datasets. Partly. Nine datasets across four asset classes and four bar sizes, each scored for realism before use (Section 3.5). Single-name equities are absent, two datasets fail the realism battery and ship with the verdict recorded, and the observation is twenty trailing closes (Section 7).

Acknowledge limitations of reused datasets. Yes. Every dataset's source, license, caveats and failing stylized facts are in Table 1 and the data scripts. The commodities file keeps a negative oil close as published and documents the opt-out.

Prepare for contamination. Yes. Sealed held-out datasets under a published hash, a canary token, and a masked view that defeats ticker memorization (Section 4.6). Forward commitment before the data exists (Section 4.5). The forward league that would exercise it has not run.

Use statistical methods. Yes. Every rank is gated on a deflated Sharpe with its number of trials, a block bootstrap with a fixed seed, and a per-run reliability test; field-wide Reality Check, SPA and step-down are reported; a bootstrap confidence interval on the DSR marks tied entries as tied. The annualization convention is stated once in Section 3.2 because getting it wrong was the first finding.

Conduct error analysis. Yes, and it is the paper. Section 5 traces every refusal to the gate that produced it and the quantity that justified it, and two of the three findings are errors in the benchmark itself.

Justify construct validity. Partly, and the gap is named. The benchmark refuses every reference agent for reasons the data makes explicit, and the luck floor never beats a reference agent. What it has

not shown is a real agent being admitted, so the claim that eligibility is attainable and the construct is not vacuous awaits the competing agent listed first in Section 7.

Statistical reporting. Every Sharpe in this paper is per period unless stated annualized, and Table 3 gives the conversion at every bar size. Every deflated Sharpe is reported beside its N . All runs use eight execution seeds and six windows; the per-cell record carries both. Costs are the typical profile of Section 3.4. Block bootstrap, not i.i.d., throughout.

Licenses. The benchmark is MIT or Apache-2.0. FRED series are public domain. Binance klines are fetched from the public endpoint under its terms and redistributed as derived daily, 4-hour, hourly and weekly closes. No dataset requires a key.

Maintenance. The benchmark is versioned on crates.io, npm and PyPI with a pinned toolchain; a tag on the main branch publishes to every registry and a verification job confirms each serves the version. Datasets are frozen by hash and regenerated only by a deliberate script run. Golden fixtures change only by a local command that refuses to run in continuous integration. Adding a dataset means adding a hash-pinned file, a fetch script with a check mode, a realism verdict, and a periods-per-year value.